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[1]: # get a single frame convert it to grayscale and pixel value of that
import cv2
import matplotlib.pyplot as plt

cap = cv2.VideoCapture("video_sample.mp4")

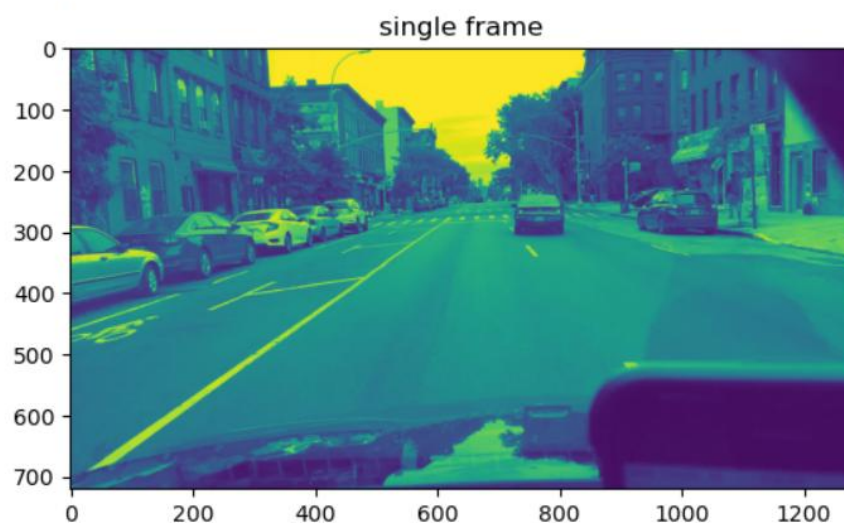
ret, img = cap.read()
gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
plt.imshow(gray_image)
plt.title("single frame")

print(gray_image)

total_frames = cap.get(cv2.CAP_PROP_FRAME_COUNT)
height = cap.get(cv2.CAP_PROP_FRAME_HEIGHT)
width = cap.get(cv2.CAP_PROP_FRAME_WIDTH)
fps = cap.get(cv2.CAP_PROP_FPS)

print(f'Total Frames: {total_frames}')
print(f'Resolution: {width}x{height}')
print(f'FPS: {fps:0.2f}')
```

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[[ 64  64  64 ... 12 12 12]
 [ 64  64  64 ... 12 12 12]
 [ 61  61  61 ... 12 12 12]
 ...
 [123 122 121 ... 34 34 34]
 [124 121 115 ... 36 36 36]
 [112 108 100 ... 36 36 36]]
Total Frames: 2398.0
Resolution: 1280.0x720.0
FPS: 59.94
```



```
[*]: import cv2

cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    cv2.imshow("GrayScale Video", gray)

    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```



```
[*]: import cv2

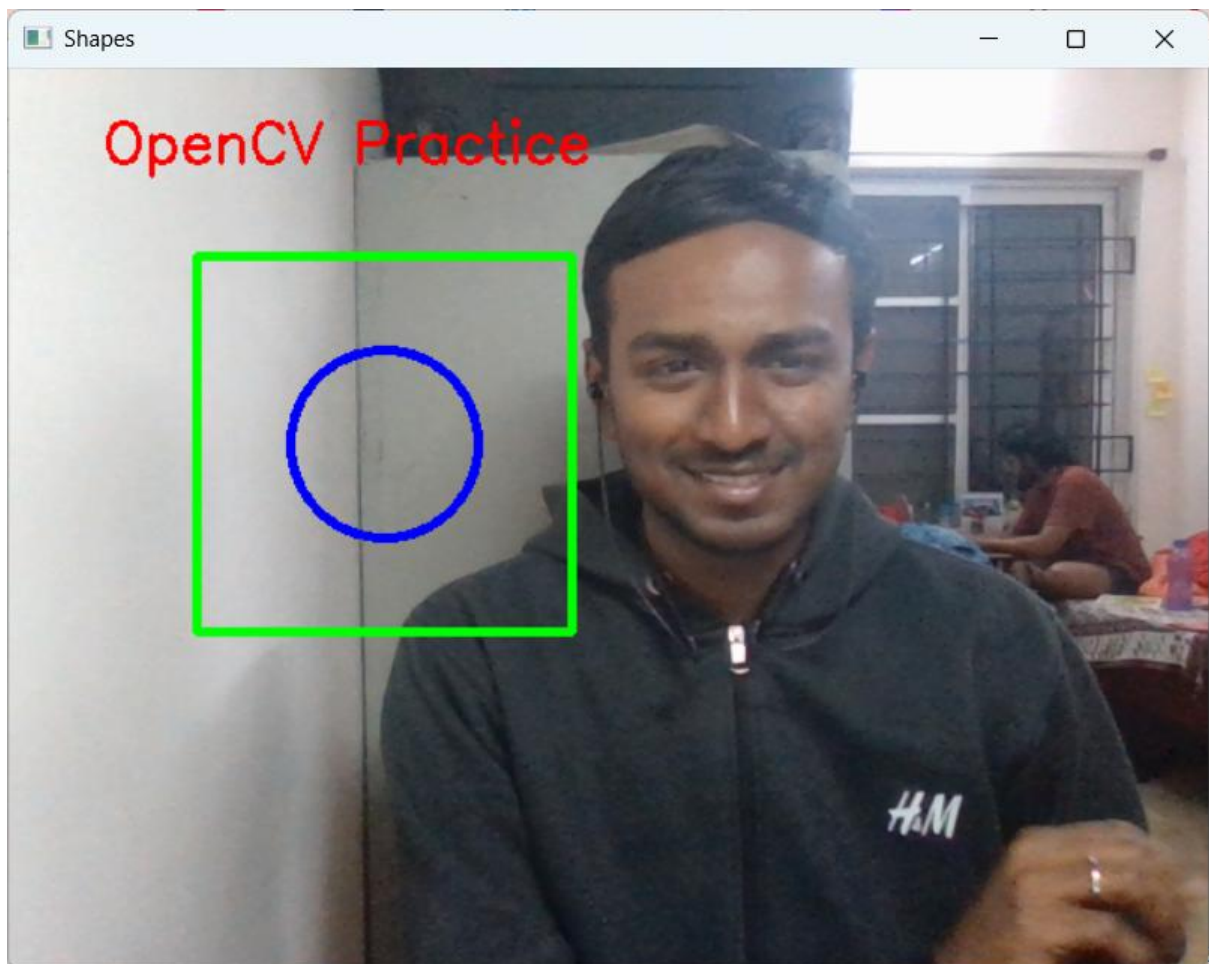
cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    cv2.rectangle(frame, (100,100), (300,300), (0,255,0), 3)
    cv2.circle(frame, (200,200), 50, (255,0,0), 3)

    cv2.putText(frame, "OpenCV Practice", (50,50), cv2.FONT_HERSHEY_SIMPLEX, 1, (0,0,255), 2)
    cv2.imshow("Shapes", frame)

    if cv2.waitKey(1) == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```



```

import cv2

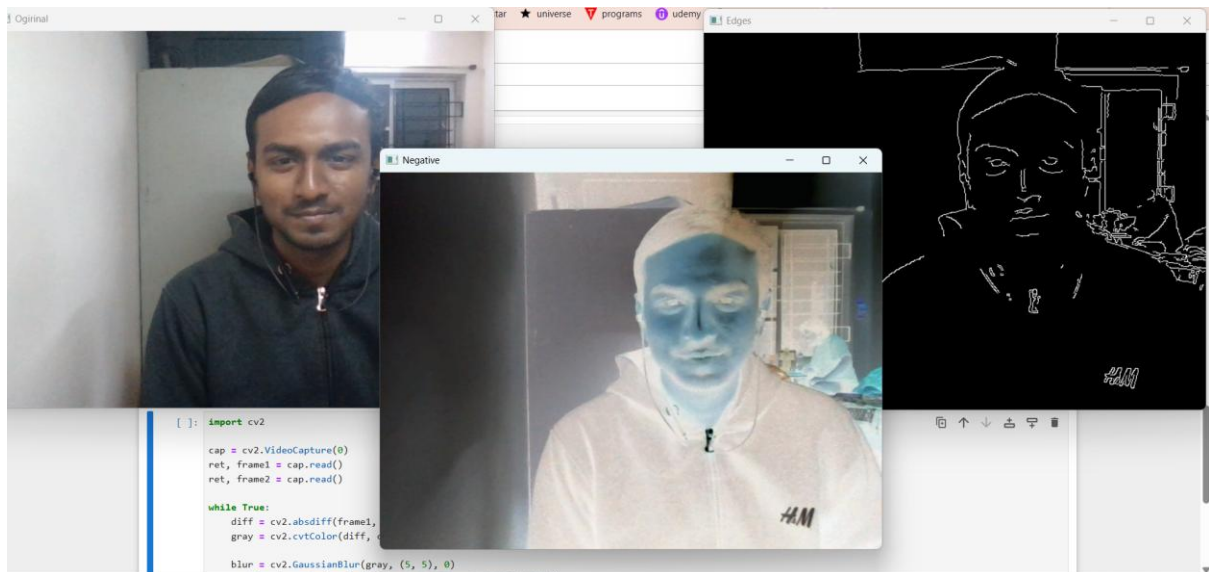
cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    edges = cv2.Canny(frame, 100, 200)
    hsv = cv2.bitwise_not(frame)
    cv2.imshow("Edges", edges)
    cv2.imshow("Original", frame)
    cv2.imshow("Negative", hsv)
    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()

```



```
[ ]: import cv2

cap = cv2.VideoCapture(0)
ret, frame1 = cap.read()
ret, frame2 = cap.read()

while True:
    diff = cv2.absdiff(frame1, frame2)
    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)

    blur = cv2.GaussianBlur(gray, (5, 5), 0)
    _, thresh = cv2.threshold(blur, 20, 255, cv2.THRESH_BINARY)

    cv2.imshow('motion', thresh)
    frame1 = frame2
    ret, frame2 = cap.read()

    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

